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ASSIGNMENT 1

Library Management System – Story Writing Workshop Report

1. Introduction

A Library Management System (LMS) is designed to automate and manage the operations of a university library. It helps users efficiently search for books, manage borrowing and returns, and maintain user accounts.

This report focuses on identifying system users (actors), developing user stories using standard agile practices, validating them using the INVEST criteria, and defining acceptance criteria for each story.

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2. Identification of Users (Actors)

In the context of a university library system, the following three primary users are identified:

2.1 Student

Students are the main users of the library system. They interact with the system to search for books, check availability, and borrow materials for academic purposes.

2.2 Librarian

Librarians are responsible for managing the day-to-day operations of the library, including issuing books, receiving returned books, and maintaining records of transactions.

2.3 Admin (System Administrator)

The admin manages the overall system, including user accounts, permissions, and system configuration to ensure smooth operation.

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3. User Stories

User stories are written using the standard agile template:

As a [user], I want [goal], so that [benefit].

3.1 User Story 1 – Student

As a **student**, I want to **search for books in the library catalog**, so that **I can find and borrow relevant materials for my studies**.

3.2 User Story 2 – Librarian

As a **librarian**, I want to **issue and receive returned books**, so that **I can manage book circulation efficiently**.

3.3 User Story 3 – Admin

As an **admin**, I want to **manage user accounts**, so that **I can control access to the library system**.

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4. Validation Using INVEST Criteria

The INVEST model ensures that user stories are well-structured and effective.

4.1 User Story 1 (Student – Search Books)

- **Independent:** The search feature can be developed separately from other features.
- **Negotiable:** Search functionality can be refined with filters (author, title, category).
- **Valuable:** Enables students to access learning materials easily.
- **Estimable:** Developers can estimate effort based on search complexity.
- **Small:** Focused only on searching functionality.
- **Testable:** Can verify if search results match keywords entered.

4.2 User Story 2 (Librarian – Issue/Return Books)

- **Independent:** Can function independently of account management.
- **Negotiable:** Workflow can be adjusted (e.g., fines, due dates).

- **Valuable:** Core functionality of library operations.
- **Estimable:** Effort can be calculated based on system requirements.
- **Small:** Focused on circulation tasks only.
- **Testable:** Can verify issuance and return operations.

4.3 User Story 3 (Admin – Manage Accounts)

- **Independent:** Does not depend on borrowing or searching features.
- **Negotiable:** Roles and permissions can be modified.
- **Valuable:** Ensures proper system access control.
- **Estimable:** Scope is clear and measurable.
- **Small:** Focused on account management tasks.
- **Testable:** Can verify account creation, updates, and deletion.

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5. Acceptance Criteria

Acceptance criteria define the conditions that must be met for a user story to be considered complete.

5.1 User Story 1 (Student – Search Books)

1. The system should display a list of books based on entered search keywords.
2. The system should show the availability status (available or borrowed) for each book.

5.2 User Story 2 (Librarian – Issue/Return Books)

1. The system should allow the librarian to issue a book and assign a due date.
2. The system should update the book status when it is returned.

5.3 User Story 3 (Admin – Manage Accounts)

1. The system should allow the admin to create, update, and delete user accounts.
2. The system should allow role assignment (student or librarian) to users.
